

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Underwater Mass Spectrometers: Some Critical Engineering Issues

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ABSTRACT

The designer of an underwater mass spectrometer faces a set of engineering trade-offs that involve power consumption, vacuum system capacity, analyte flow, time response, mass range, resolution, precision, detection limits, and depth capability. In this paper we first discuss a set of engineering approaches and models that have been found helpful in the design of the NEREUS system, and are believed to be generally applicable to other underwater instruments. Secondly, we focus on the design of sample inlet systems, and on means of limiting the power consumption of a mass spectrometer that must operate for extended periods on battery power. As one example, detailed design of the highly-efficient emission regulator used on NEREUS and applicable to many instruments using EI ionization will be presented.